

121. If the foci of a hyperbola are same as that of the ellipse $\frac{x^2}{9} + \frac{y^2}{25} = 1$ and the eccentricity of the hyperbola is $\frac{15}{8}$ times the eccentricity of the ellipse, then the smaller focal distance of the point $\left(\sqrt{2}, \frac{14}{3}\sqrt{\frac{2}{5}}\right)$ on the hyperbola, is equal to :

यदि एक अतिपरवलय की नाभियाँ दीर्घवृत्त $\frac{x^2}{9} + \frac{y^2}{25} = 1$ की नाभियों के समान हैं तथा अतिपरवलय की उत्केन्द्रता, दीर्घवृत्त की

उत्केन्द्रता का $\frac{15}{8}$ गुना है, तो अतिपरवलय पर बिन्दु $\left(\sqrt{2}, \frac{14}{3}\sqrt{\frac{2}{5}}\right)$ की छोटी नाभीय दूरी बराबर है :

- (A) $7\sqrt{\frac{2}{5}} + \frac{8}{3}$ (B) $7\sqrt{\frac{2}{5}} - \frac{8}{3}$ (C) $14\sqrt{\frac{2}{5}} - \frac{4}{3}$ (D) $14\sqrt{\frac{2}{5}} - \frac{16}{3}$

Ans. B

Sol. $\frac{x^2}{9} + \frac{y^2}{25} = 1$

$a = 3, b = 5,$

$e = \sqrt{1 - \frac{9}{25}} = \frac{4}{5}$

foci = $(0, \pm be) = (0, \pm 4)$

Eccentricity of hyperbola $e_H = \frac{4}{5} \times \frac{15}{8} = \frac{3}{2}$

Let equation of hyperbola be

$\frac{x^2}{A^2} - \frac{y^2}{B^2} = -1$

According to equation $Be_H = 4$

$\therefore B = \frac{8}{3}$

$\therefore A^2 = B^2 (e_H^2 - 1)$

$= \frac{64}{9} \left(\frac{9}{4} - 1 \right) \Rightarrow A^2 = \frac{80}{9}$

$\therefore \frac{x^2}{\frac{80}{9}} - \frac{y^2}{\frac{64}{9}} = -1$

$$\text{Directrix } y = \pm \frac{B}{e_H} = \pm \frac{16}{9}$$

$$PS = ePM$$

$$= \frac{3}{2} \left| \frac{14}{3} \sqrt{\frac{2}{5}} - \frac{16}{9} \right|$$

$$= 7 \sqrt{\frac{2}{5}} - \frac{8}{3}$$

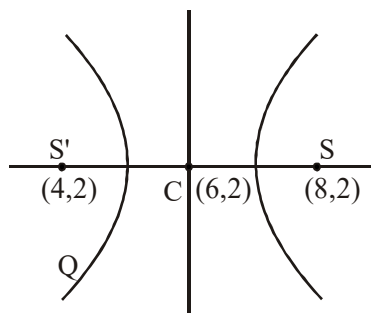
122. If the equation of the hyperbola with foci (4, 2) and (8, 2) is $3x^2 - y^2 - \alpha x + \beta y + \gamma = 0$, then $\alpha + \beta + \gamma$ is equal to _____.

यदि नाभियों (4, 2) और (8, 2) के अतिपरवलय का समीकरण $3x^2 - y^2 - \alpha x + \beta y + \gamma = 0$ है, तो $\alpha + \beta + \gamma$ बराबर है

_____.

Ans. 141

Sol.



Equation of hyperbola is

$$\frac{(x-6)^2}{a^2} - \frac{(y-2)^2}{4-a^2} = 1$$

$$\Rightarrow (4-a^2)(x-6)^2 - a^2(y-2)^2 = a^2(4-a^2)$$

Comparing with $3x^2 - y^2 - \alpha x + \beta y + \gamma = 0$, we get

$$a^2 = 1 \text{ and } \alpha = 36, \beta = 4 \text{ and } \gamma = 101$$

$$\therefore \alpha + \beta + \gamma = 141$$

123. Consider the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ having one of its focus at $P(-3, 0)$. If the latus rectum through its other focus subtends a right angle at P and $a^2 + b^2 = \alpha\sqrt{2} - \beta$, $\alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is _____.

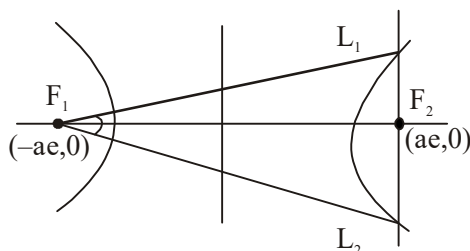
अतिपरवलय $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, जिसकी एक नाभि $P(-3, 0)$ पर है, का विचार कीजिए। यदि इसकी दूसरी नाभि से होकर जाने वाली

नाभिलम्ब जीवा, बिन्दु P पर एक समकोण बनाती है और $a^2 + b^2 = \alpha\sqrt{2} - \beta$, $\alpha, \beta \in \mathbb{N}$ हैं, तो $\alpha + \beta$ बराबर है _____.

Ans. 1944

Sol. $f_1 \equiv (-a, e, 0) \equiv P(-3, 0)$

$$\Rightarrow ae = 3$$



$$\tan 45^\circ = \frac{b^2}{2ae}$$

$$2ae = \frac{b^2}{a}$$

$$b^2 = 6a$$

$$\text{Also } a^2e^2 = a^2 + b^2$$

$$9 = a^2 + 6a$$

$$a^2 + 6a - 9 = 0$$

$$a = -3 \pm 3\sqrt{2} = -3(1 \pm \sqrt{2})$$

$$\therefore a^2b^2 = a^2 \cdot 6a = 6a^3$$

$$= 6(135\sqrt{2} - 189)$$

$$\alpha = 810 \text{ and } \beta = 1134$$

$$\therefore \alpha + \beta = 1944$$

124. Let the circle C touch the line $x - y + 1 = 0$, have the centre on the positive x-axis, and cut off a chord of length

$\frac{4}{\sqrt{13}}$ along the line $-3x + 2y = 1$. Let H be the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$, whose one of the foci is the centre of C and the length of the transverse axis is the diameter of C. Then $2\alpha^2 + 3\beta^2$ is equal to _____.

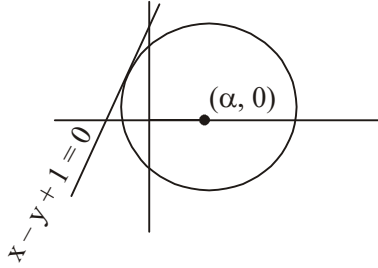
माना कि वृत्त C, रेखा $x - y + 1 = 0$ को स्पर्श करता है, इसका केन्द्र धनात्मक x-अक्ष पर है और यह रेखा $-3x + 2y = 1$ के अनुदिश

लम्बाई $\frac{4}{\sqrt{13}}$ की एक जीवा काटता है। माना कि H अतिपरवलय $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ है, जिसकी एक नाभि, C का केन्द्र है और अनुप्रस्थ

अक्ष की लम्बाई, C का व्यास है। तब $2\alpha^2 + 3\beta^2$ बराबर है _____.

Ans. 19

Sol.



$$\left| \frac{\alpha + 1}{\sqrt{2}} \right| = r$$

$$p = \left| \frac{-3\alpha - 1}{\sqrt{13}} \right| = \frac{3\alpha + 1}{\sqrt{13}}$$

Focus of hyperbola (3, 0)

$$\frac{4}{\sqrt{13}} = 2\sqrt{r^2 - p^2}$$

$$ae = 3, 2a = 4\sqrt{2}$$

$$\frac{4}{\sqrt{13}} = 2\sqrt{\frac{(\alpha + 1)^2}{2} - \frac{(3\alpha + 1)^2}{13}}$$

$$e = \frac{3}{2\sqrt{2}}, a = 2\sqrt{2}$$

$$2\sqrt{2} = \sqrt{13(\alpha + 1)^2 - 2(3\alpha + 1)^2}$$

$$e^2 = 1 + \frac{b^2}{a^2}$$

$$13\alpha^2 + 26\alpha + 13 - 18\alpha^2 - 12\alpha - 2 = 8$$

$$\frac{9}{8} = 1 + \frac{b^2}{8}$$

$$-5\alpha^2 + 14\alpha + 3 = 0$$

$$b^2 = 1$$

$$5\alpha^2 - 14\alpha - 3 = 0$$

$$2\alpha^2 + 3\beta^2 = 16 + 3 = 19$$

$$5\alpha^2 - 15\alpha + \alpha - 3 = 0$$

$$5\alpha(\alpha - 3) + 1(\alpha - 3) = 0$$

$$\alpha = 3, \alpha = -\frac{1}{5} \text{ (rejected)}$$

125. Let $H_1: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $H_2: -\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$ be two hyperbolas having length of latus rectums $15\sqrt{2}$ and $12\sqrt{5}$ respectively. Let their eccentricities be $e_1 = \sqrt{\frac{5}{2}}$ and e_2 respectively. If the product of the lengths of their transverse axes is $100\sqrt{10}$, then $25e_2^2$ is equal to _____.

माना कि $H_1: \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ और $H_2: -\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1$ दो अतिपरवलय हैं, जिनकी नाभिलम्ब जीवाओं की लम्बाइयाँ क्रमशः

$15\sqrt{2}$ और $12\sqrt{5}$ हैं। माना कि उनकी उत्केन्द्रताएँ क्रमशः $e_1 = \sqrt{\frac{5}{2}}$ और e_2 हैं। यदि उनके अनुप्रस्थ अक्षों की लम्बाइयों का

गुणनफल $100\sqrt{10}$ है, तो $25e_2^2$ बराबर है _____.

Ans. 55

Sol. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$; LLR = $\frac{2b^2}{a} = 15\sqrt{2}$ _____(1)

$-\frac{x^2}{A^2} + \frac{y^2}{B^2} = 1 \Rightarrow \text{LLR} \Rightarrow \frac{2A^2}{B} = 12\sqrt{5}$ _____(2)

$e_1 = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{5}{2}} \Rightarrow \frac{b^2}{a^2} = \frac{3}{2}$

from (1) $3a = 15\sqrt{2}$ $a = 5\sqrt{2}$
 $b^2 = 75$

$b = 5\sqrt{3}$

$\Rightarrow (2a)(2B) = 100\sqrt{10}$

$2B = 10\sqrt{5}$

$B = 5\sqrt{5}$

from (2) $\Rightarrow 2A^2 = 300$
 $A^2 = 150$

$25e_2^2 = 25\left(1 + \frac{150}{125}\right) = 55$

126. The length of the latus rectum and directrices of a hyperbola with eccentricity e are 9 and $x = \pm \frac{4}{\sqrt{3}}$, respectively.

Let the line $y - \sqrt{3}x + \sqrt{3} = 0$ touch this hyperbola at (x_0, y_0) . If m is the product of the focal distances of the point (x_0, y_0) , then $4e^2 + m$ is equal to _____.

किसी e उत्केद्रता वाले अतिपरवलय, जिसके नाभीलंब की लम्बाई तथा नियताएँ क्रमशः 9 तथा $x = \pm \frac{4}{\sqrt{3}}$ हैं। माना रेखा

$y - \sqrt{3}x + \sqrt{3} = 0$ इस अतिपरवलय को (x_0, y_0) पर स्पर्श करती है। यदि m बिन्दु (x_0, y_0) की नाभीय दूरियों का गुणनफल है, तो $4e^2 + m$ बराबर है _____

Ans. 61

Sol. $\frac{2b^2}{a} = 9$ and $\pm \frac{a}{e} = \pm \frac{4}{\sqrt{3}}$ (1)

Equation of tangent $y = \sqrt{3}x - \sqrt{3}$

$m = \sqrt{3}$

$c = -\sqrt{3}$

Condition of tangency $c = \pm \sqrt{a^2 m^2 - b^2}$

$$-\sqrt{3} = \sqrt{3a^2 - b^2}$$

$$6a^2 - 9a - 6 = 0$$

$$2a^2 - 3a - 2 = 0$$

$$2a^2 - 4a + a - 2 = 0$$

$$(a - 2)(2a + 1) = 0$$

$$a = 2 \text{ or } -\frac{1}{2}$$

When $a = 2$, $b^2 = 9$ put $a = 2$ in(1)

$$\frac{2}{e} = \frac{1}{\sqrt{3}} \Rightarrow e = \frac{\sqrt{3}}{2}$$

Which is impossible

So question is bonus.

127. Let S be the focus of the hyperbola $\frac{x^2}{3} - \frac{y^2}{5} = 1$, on the positive x-axis. Let C be the circle with its centre at $A(\sqrt{6}, \sqrt{5})$ and passing through the point S. If O is the origin and SAB is a diameter of C, then the square of the area of the triangle OSB is equal to _____.

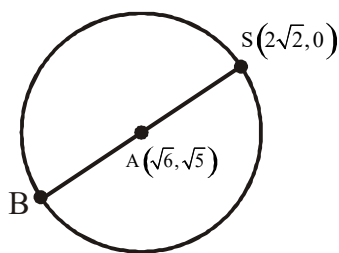
माना अतिपरवलय $\frac{x^2}{3} - \frac{y^2}{5} = 1$ की धनात्मक x-अक्ष पर नाभि S है। माना बिंदु S से होकर जाने वाले एक वृत्त C का केन्द्र $A(\sqrt{6}, \sqrt{5})$ है। यदि O मूलबिन्दु है तथा C का एक व्यास SAB है, तो त्रिभुज OSB के क्षेत्रफल का वर्ग है _____.

Ans. 40

Sol. $\frac{x^2}{3} - \frac{y^2}{5} = 1$

$$5 = 3(e^2 - 1) \Rightarrow e = \sqrt{\frac{8}{3}}$$

$$S \equiv (2\sqrt{2}, 0)$$



A is mid-point of BS

$$\Rightarrow B(2\sqrt{6} - 2\sqrt{2}, 2\sqrt{5})$$

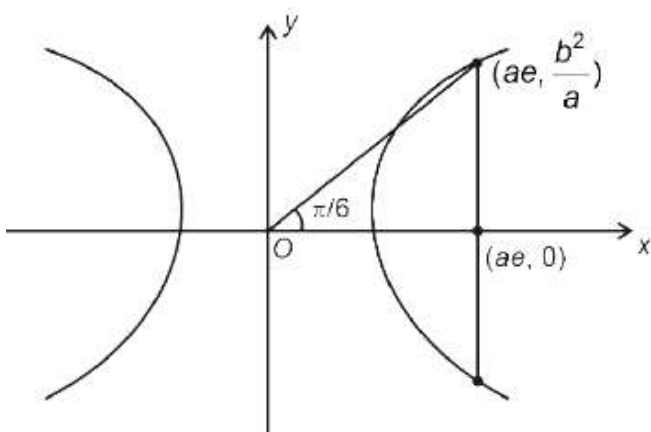
$$\Delta(OSB) = \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 2\sqrt{2} & 0 & 1 \\ 2\sqrt{6} - 2\sqrt{2} & 2\sqrt{5} & 1 \end{vmatrix} = 2\sqrt{10}$$

$$(\Delta(OSB))^2 = 40$$

128. Let the latus rectum of the hyperbola $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ subtend an angle of $\frac{\pi}{3}$ at the centre of the hyperbola. If b^2 is equal to $\frac{\ell}{m}(1 + \sqrt{n})$, where ℓ and m are co-prime numbers, then $\ell^2 + m^2 + n^2$ is equal to _____.

माना अतिपरवलय $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ की नाभिलम्ब जीवा, अतिपरवलय के केन्द्र पर $\frac{\pi}{3}$ का कोण बनाती है। यदि $b^2 = \frac{\ell}{m}(1 + \sqrt{n})$ है, जहाँ ℓ तथा m असहभाज्य संख्याएँ हैं, तो $\ell^2 + m^2 + n^2$ बराबर है _____।

Ans. 182



Sol.

$$\Rightarrow \tan \frac{\pi}{6} = \frac{b^2}{a^2 e}$$

$$\therefore \frac{b^2}{a^2} = (e^2 - 1)$$

$$\frac{1}{\sqrt{3}} = \frac{e^2 - 1}{e} \Rightarrow \sqrt{3}e^2 - e - \sqrt{3} = 0$$

$$e = \frac{1 \pm \sqrt{1+12}}{2\sqrt{3}}, \because e > 1 \Rightarrow e = \frac{\sqrt{13}+1}{2\sqrt{3}}$$

$$\Rightarrow b^2 = 9 \left(\frac{13+1+2\sqrt{13}}{12} - 1 \right) = 9 \left(\frac{2+2\sqrt{13}}{12} \right)$$

$$\Rightarrow b^2 = 9 \left(\frac{1+\sqrt{13}}{6} \right)$$

$$b^2 = \frac{3}{2} (1 + \sqrt{13})$$

$$\Rightarrow \ell = 3, m = 2, n = 13$$

$$\Rightarrow \ell^2 + m^2 + n^2 = 182$$